# Math 534 H - Spring 2026 - Lecture 14

We continue with the proof of the (weak) Maximum Principle.

2 We refine the first approach from last time to make it successful. We do this via a PERTURBATION ARGUMENT:

Define a new function:

$$v(x,t) := u(x,t) + \varepsilon x^2 \quad , \quad \varepsilon > 0$$

arbitrary, to be taken as small as we need to.

Let $M := \max_{(x,t)\in \Gamma} \{u(x,t)\}$

Recall <u>WTS</u> $\max_{(x,t)\in R} \{u(x,t)\} = M$ (*)

$R = [0,L]\times[0,T]$

<u>NOTE</u> $\Gamma \subseteq R$ then $\max_{(x,t)\in R} \{u(x,t)\} \geq M$ ✓

To show (*) we only need to prove

GOAL $\max_{(x,t)\in R} \{u(x,t)\} \leq M$ ⟸ (†)

$\left( \Longleftrightarrow \quad \underline{u(x,t) \leq M} \ \underline{\text{for all}} \ \underline{(x,t)\in R} \right)$

Now from the definition of $v$ and of $M$ we know that for all $(x,t) \in \Gamma$

$$\boxed{v(x,t) \leq M + \varepsilon L^2} \quad (**)$$

CLAIM: Our goal $(\dagger)$ follows from proving that $v(x,t) \leq M + \varepsilon L^2$ for all $(x,t) \in R$

Indeed if the latter is true then by definition of $v(x,t)$ we get that for $(x,t) \in R$ $u(x,t) \leq M + \varepsilon(L^2 - x^2)$

And since $x \in [0,L] \Rightarrow \underbrace{L^2 - x^2}_{0 \leq \cdot \leq L^2} \geq 0$ and bounded

and so by letting $\varepsilon \to 0^+$ we obtain our goal for $u(x,t)$ (†).

Back to (**) :

$$V_t - k V_{xx} = \underbrace{(u_t - k u_{xx})}_{=0} - 2k\varepsilon .$$

(*)

$$\therefore \quad \boxed{V_t - k V_{xx} = -2k\varepsilon < 0 , \text{ on } R}$$

rectangle.

Now let's repeat the argument on our first approach applied to $V$ instead of $u$.

Suppose $v(x,t)$ attains its maximum at an interior point $(x_0, t_0)$ of $R$. Then necessarily $v_t(x_0,t_0) = 0$ $v_{xx}(x_0,t_0) \leq 0$

Hence at $(x_0,t_0)$ $\underbrace{v_t(x_0,t_0)}_{=0} - k\underbrace{v_{xx}(x_0,t_0)}_{\leq 0} \geq 0$

This contradicts $\circledast$

Hence $v$ cannot attain a maximum in the interior of $R$

How about the fourth side when $\boxed{t_0 = T}$

Then if you have a maximum at $(x_0, \underline{T})$

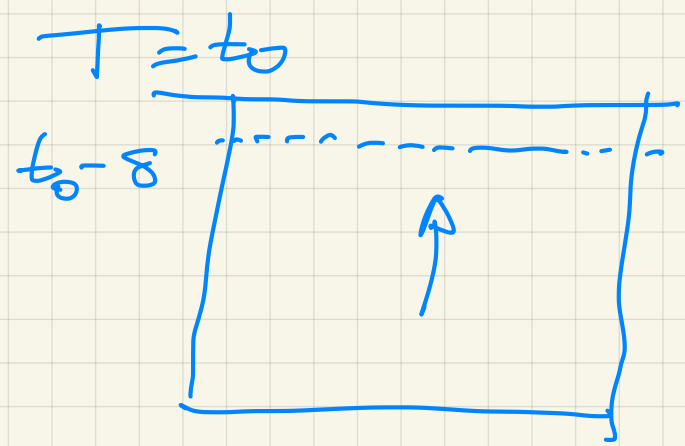
Then we still have that $V_x(x_0, T) = 0$ and $V_{xx}(x_0, \underline{t_0}) \leq 0$. But $V_t(x_0, \underline{t_0})$ does <u>NOT</u> have to be zero at $t_0 = T$:

<u>indeed</u>

$$V_t(x_0, t_0) = \lim_{\boxed{\delta \to 0^+}} \frac{V(x_0, t_0) - V(x_0, t_0 - \delta)}{\delta}$$

But if $V$ attains a max at $(x_0, t_0)$ then

$$V(x_0, t_0) \geq V(x_0, t_0 - \delta)$$




Hence $V_t(x_0, t_0) \geq 0$

Hence $\underbrace{v_t(x_0, t_0)}_{\geq 0} - k \underbrace{v_{xx}(x_0, t_0)}_{\leq 0} \geq 0$

$t_0 = T$

which contradicts again (*)

This proves that

we just proved this

$$v(x,t) \leq \max_{(x,t)\in R} \{v(x,t)\} = \max_{(x,t)\in \Gamma} \{v(x,t)\}$$

$$\leq M + \varepsilon L^2$$

which proves (**)

## UNIQUENESS AND STABILITY of solutions to the heat/diffusion equation

We use <u>max/min principle</u>.

Consider

(IVBP) for heat Dirichlet Boundary Conditions

$$\begin{cases} u_t - k\, u_{xx} = 0 & 0 \le x \le L \quad t > 0 \\ u(x,0) = \phi(x) & x \in [0, L] \\ u(0,t) = g(t), \quad u(L,t) = h(t) & t > 0 \end{cases}$$

How do we prove uniqueness: Assume that we have $u_1$ and $u_2$ both solution to <u>the SAME</u> (IVBP)

Let $w = u_1 - u_2$ then $w$ satisfies

$$\begin{cases} w_t - k w_{xx} = 0 & x \in [0,L] \quad t > 0 \quad (0 < t \le T) \\ w(x,0) = 0 & x \in [0,L] \\ w(0,t) = 0 \; ; \; w(L,t) = 0, & t > 0 \end{cases}$$

But

$$\min_{(x,t)\in R} \{w(x,t)\} \le w(x,t) \le \max_{(x,t)\in R} \{w(x,t)\}$$

By max/min principle

BY MAX/MIN PPLE

$$\min_{(x,t)\in \Gamma} \{w(x,t)\} = 0$$

By MAX/MIN PPLE

$$\max_{(x,t)\in \Gamma} \{w(x,t)\} = 0$$

$$\Gamma = \{(x,t) \,/\, t = 0 ; x = 0 \text{ or } x = L\}$$

Hence $0 \leq w(x,t) \leq 0$ for all $(x,t) \in R$

$(R = [0, L] \times [0, T])$

So $w(x,t) = u_1(x,t) - u_2(x,t) = 0 \ \forall (x,t) \in R$

$\Rightarrow \quad u_1(x,t) = u_2(x,t) \ \forall (x,t) \in R$

$\Rightarrow$ we have uniqueness as desired. #.